What matters in an LLM forecasting prompt

One-page synthesis of the SaferAI forecaster experiments, Spring–August 2026 · Madhav Khanal, with Jakub Kryś, Jeff Mohl, Matt Smith · 2026-08-30

1 Task, data, instruments

A forecaster LLM (Claude Sonnet 4.6) predicts $p = \Pr[\text{model } m \text{ solves task } t \text{ in one attempt}]$ for cells (t, m) of the Lyptus cyber-benchmark outcome matrix. Its evidence: example tasks with solved/failed outcomes and group pass rates from *other* difficulty bins; the target's own FST difficulty bin is never shown. **Accuracy**: Brier $(p_{50} - y)^2$; CRPS of the Beta fitted to (p_{25}, p_{50}, p_{75}). **Output shift**: Wasserstein-1 (W_1) between p_{50} distributions. **Sets** (frozen seed-42 split of 5 train benchmark families): $D_{\text{search}} = 84$ cells (selection), $D_{\text{held}} = 230$ cells (21 unseen tasks × 11 models); CVEBench+CyberGym reserved. Re-measurement noise: sd 0.0065 per single D_{search} pass, ≈ 0.002 per multi-pass D_{held} estimate. **Reference points on D_{held}**: hand-written minimal `seed` prompt 0.1306; *difficulty-oracle table* — predict m's leave-one-out mean pass rate over the other tasks in t's FST bin, a lookup using the very label the LLM is denied — 0.1034.

2 Where the 82 prompts came from, and what was already known

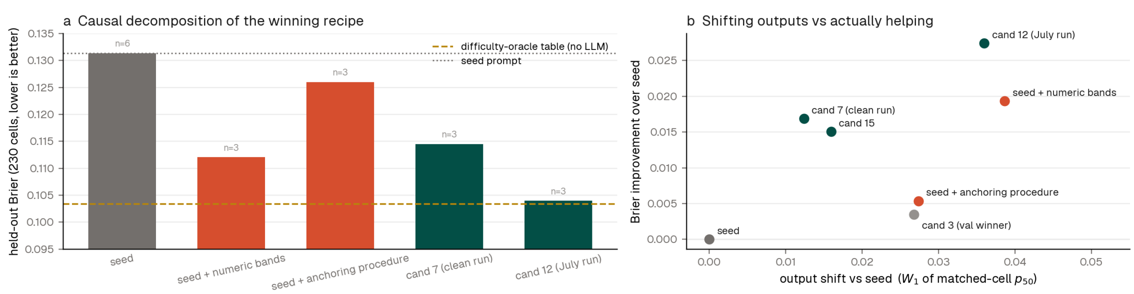
GEPA (reflective evolutionary prompt search, arXiv:2507.19457) ran twice under identical configs — July, and a clean re-run after fixing a scoring bug (unparsable responses scored Brier 1.0; now retried, then halt; 0 failures in 8,300 calls) — plus two search-setting ablations: **82 evolved prompts**, 13 evaluated on D_{held} (7 multi-pass). Both main runs found genuine held-out winners (`cand12` +0.027, better than seed in 8/8 paired passes; `cand7` +0.016, 3/3); both times the run's *own* D_{search} winner was $\approx$ null on D_{held}, so final selection must use held-out multi-pass scores, never the search set. Prior context: hand-ablations of the original pipeline prompt (Exp. I, 300 matched cells) put the whole scaffold at 0.014 Brier — only the stage-1 analysis call individually significant (+0.0105); evidence *texts* load-bearing (+0.0178 removed, p=10^{-7}); pass-rate numbers and solved/failed tags free riders (n.s.). Exp. II: the forecaster is already calibrated (ECE 0.064, recalibration moves < 0.003), so any prompt gain must buy *discrimination*, not calibration.

3 Feature analysis: tag every prompt, screen on two axes, then intervene

Each evolved prompt was tagged by a deterministic rubric with five pre-registered features; each is screened for **accuracy** (ΔBrier on D_{search}, within-run then pooled — runs differ in temperature) and **output shift** (W_1 vs the same run's seed on identical cells; all features shift 0.023–0.028 because prompts carry feature *bundles*). The sealed-null v2 run separates the bundle observationally: its prompts carry anti-hedging + task-typing + anchoring language but *no* numeric bands or nearest-example rules, and gained nothing.

Feature (of 82 evolved prompts)	prevalence	ΔBrier, D_{search}	reading
reference-class anchoring instruction	82/82	—	universal: the optimizer's first move in every lineage
nearest-example weighting ("near-twin")	31	−0.0043	strongest discriminator; in winners, absent in v2 nulls
numeric probability bands	45	−0.0008	in all 3 July winners, absent in v2 nulls; **causal, below**
anti-hedging (ban the 0.5 default)	46	−0.0025	in all winners *and* all v2 nulls: insufficient alone
task-type rules (interactive vs. static)	69	−0.0029	near-universal; insufficient alone (v2)

Causal check (3 paired D_{held} passes, same-session seed): re-injecting the winners' two halves into the seed separately gives *seed+numeric bands* 0.1120 (paired +0.020, better in 3/3 passes) versus *seed+anchoring procedure* 0.1260 (paired +0.006, 3/3). The decomposition is one-sided: the explicit numeric anchors alone recover ~74% of the full winner's edge; the reference-class procedure alone is real but 3× smaller. This retro-explains the v2 null (its instruction banned numbers) and names the mechanism: *the forecaster follows anchors it is given; it does not construct equivalent anchors from evidence, even when instructed step-by-step.* The full winner (+0.027) still beats bands alone: the procedure adds value only on top of explicit numbers. Beta-CRPS agrees everywhere (seed 0.216, bands 0.189, `cand12` 0.178).



4 Conclusions

(i) Anchoring is the master variable at every layer of this project. In the spring risk-elicitation study (different task: risk-model parameter beliefs), outputs tracked any *provided* number nearly linearly; removing the anchor exploded variance (Fréchet-mean W_1=0.154, 18×), personas and scaffolding inert — anchoring as *bias risk*. In forecasting, the model barely exploits evidence numbers spontaneously (Exp. I); GEPA instructed anchoring in 82/82 evolved prompts, its winning recipe being reference-class forecasting (anchor on the closest analog group's empirical rate, correct toward near-twins) — anchoring turned into the *mechanism*. **(ii) The gain is resolution, exactly as Exp. II demanded:** `cand12` vs seed decomposes as resolution 0.078→0.110, reliability unchanged; Exp. II's open question is closed at ≈ 0.03 Brier of extractable discrimination. **(iii) Two independent levers hit the same ceiling.** Optimized Sonnet reaches the difficulty-oracle table (0.104 vs 0.103) while blind to the difficulty label; on the spring grid the best base model (GPT-5.5, hand prompt) tied the same oracle family (0.102 vs bin-table 0.098, IRT 0.093) where Sonnet lagged (0.124). Prompt evolution is worth about one model generation, and both levers saturate at the oracle line — the practical ceiling given single-run ground truth. **(iv) What remains**: the reserved one-shot transfer test (CVEBench+CyberGym) — out of distribution, where the oracle table must extrapolate too, does the recipe keep its edge?

Reproducibility. Scripts `feature_report_data.py`, `val_noise_study.py`, `parse_failure_audit.py` (gepa repo); raw runs on `results/*` branches; curated data, prompt texts, this document: `experiments/III_gepa_optimization/summary/`; spring sources `prompt_sensitivity/`, Exp. I/II, final report §3.